

Education

Massachusetts Institute of Technology

Cambridge, MA

Bachelor of Science in **Computer Science and Engineering** (Course 6-3)

Sep. 2022 - May 2025

Bachelor of Science in **Mathematics** (Course 18)**GPA: 5.0/5.0**

Graduate coursework: Multiagent Learning, Sensorimotor Learning, Symmetry Machine Learning, Natural Language Processing, High-Dimensional Statistics, Stochastic Processes, Quantum Computation, Programming Synthesis, Secure Hardware Design

Sharif University of Technology

Tehran, Iran

Transferred to MIT after second year; **GPA: 3.9/4.0**

Sep. 2020 - May 2022

Publications

Probabilistic Homotopy Optimization for Dynamic Motion Planning

published in IROS 2024

Shayan Pardis*, Matthew Chignoli*, Sangbae Kim

<https://arxiv.org/abs/2408.12490>

Research Experience

Diffusion-Based Uncertainty Guidance for Video-Language Planning

Cambridge, MA

MIT; Supervised by Prof. Leslie Kaelbling and Prof. Yilun Du

Sep. 2024 - May 2025

- Developed a diffusion-based planning framework using ELBO as a proxy for uncertainty, guiding model rollouts to select high-likelihood actions across the joint distribution of vision and action models.
- Empirically showed that the standard deviation of ELBO approximates the curvature of the log likelihood (Hessian), and used this for uncertainty-guided denoising of the action model.

Homotopy Optimization for Acrobatic Humanoid Motion Planning

Cambridge, MA

MIT; Supervised by Prof. Sangbae Kim

Feb. 2023 - May 2024

- Designed an optimization method inspired by Curriculum Learning and Probabilistic Roadmaps (PRM) that traverses the multi dimensional homotopy space from a relaxed (easy) problem to the original problem. This framework automates the discovery of highly dynamic trajectories (e.g. acrobatic maneuvers for humanoids), which previously required handcrafted expert initialization and heuristics. Published in IROS 2024.
- Enhanced the throughput of the QP-based controller for the MIT Humanoid robot by 4x through parallel programming techniques.

Awards

Gold medal (rank 10) in International Olympiad in Informatics (2020)

ICPC 2021 World Finalist (Asia-Tehran region champion)

Gold medal (rank 1) in Iran National Olympiad in Informatics (2019)

Silver medal (rank 24) in Asia-Pacific Informatics Olympiad (2020)

Work Experience

Cursor

San Francisco, CA

ML Research Engineer

Jun. 2025 - Present

- Developed a value-model based RL algorithm inspired by VAPO that improves long-horizon credit assignment over GRPO, yielding higher token efficiency and more stable training dynamics.
- Implemented Expert Parallelism with overlapping compute/communication, yielding a 30% faster MoE forward pass and a 50% reduction in activation memory for SwiGLU layers.
- Built the Autoinstall pipeline to process raw codebases into runnable tasks for RL training, performing dependency resolution, package installation, and runnability checks within the existing sandbox system.

Backflip AI

San Francisco, CA

ML Research/Engineering Intern

Jan. 2025 - Jan. 2025

- Trained Vision Transformer models to convert hand-drawn sketches into parameterized shapes, a preliminary step for a 3D CAD editing tool.
- Designed a GAN-inspired architecture to capture the symmetry of solutions.

Citadel LLC

New York City, NY

Quantitative Developer Intern at Central Risk Engineering

Jun. 2024 - Aug. 2024

- Developed distributed system infrastructure tools utilizing Kubernetes, gRPC, multiprocessing, Cloud Run, Redis; **Secured a return offer.**
- Designed a parallel testing framework that identified performance bottlenecks, achieving a 2x speedup by optimizing the queuing mechanism.

Google Summer of Code <i>Julia CUDA Developer</i> - Developed CUDA kernels that enable simulation of Quantum Clifford gates; under the supervision of Prof. Stefan Krastanov - Contributed to QuantumClifford.jl, a Julia package for Quantum Error-Correcting Codes; achieved 10x speedup (details)	Mountain View, CA (Remote) Jun. 2023 - Sep. 2023
SIMCON <i>Geometric Algorithm Design Intern</i> Designed a 3D mesh contraction algorithm to convert meshes into skeleton graphs with enhanced accuracy and 2x speedup (in C++)	Wuerselen, Germany (Remote) Sep. 2021 - Mar. 2022
Carriot <i>Data Science Intern</i> Designed and trained a model to map addresses to their corresponding locations (geocoding problem) utilizing OSM and Elasticsearch.	Tehran, Iran Jul. 2021 - Sep. 2021
Abarkelas <i>Web Developer (Part-Time)</i> Developed backend (Django) and frontend (NuxtJs). Set up Prometheus and Grafana for monitoring. Created PWA for the website	Tehran, Iran Oct. 2020 - Jun. 2021
Projects	
Novel Shape Generation with SO3-Equivariant Auto-Encoders (MIT 6.S966, Symmetry ML)  Repository Designed an SO(3) equivariant autoencoder using spherical harmonics and a latent space traversal that separates rotation from deformation.	Apr. 2024 - May 2024
Decision State Space Models (MIT 6.8200, Sensorimotor Learning)  Repository Reimplemented Decision Transformer replacing transformer with S4 model and demonstrated improved performance in credit assignment tasks.	Apr. 2024 - May 2024
Formal Complexity Verification (MIT 6.S981, Programming Synthesis)  Repository Formulated time complexity verification of a program as synthesizing a fix-point function. The demo uses a custom language with Python syntax.	Oct. 2023 - Dec. 2023
FaceExplore (Personal Project, to be used at MIT Ring Delivery)  Repository Developed a face search engine using a pretrained ResNet for feature extraction and a custom hierarchical clustering method to identify faces of over 500 students in a large (40GB) dataset of images. Utilized MTCNN for face detection and React, Flask, Nginx, and Docker for the website.	Jun. 2023 - Aug. 2023
Sharif AI Challenge  Repository Developed AI agents for a multiagent game, achieving 4th place in the competition. Used Huffman code for cost-efficient communication and A* algorithm for planning under uncertainty.	Mar. 2021 - May 2021
ShelveBot (MIT 6.4210, Robotics Manipulation)  Repository Demonstrated pick-and-place capabilities on the PR2 robot. Integrated RRT, Inverse Kinematics, and Trajectory Optimization for motion planning. Utilized a custom DSL for symbolic planning and optimized grasp selection. Recieved the Outstanding Project Award .	Oct. 2023 - Dec. 2023
Teaching and Leadership	
Author of Olympiad Graph Theory Book Led a team of 10 peers to write a 200-page online book on graph theory in Persian, available at gtol.shaaazzz.ir , focusing on algorithmic approaches to graph theory concepts which helped thousands of students prepare for the Iranian National Olympiad in Informatics.	Feb. 2020 - Dec. 2021
Algorithm Course Organizer (Iranian National Olympiad in Informatics Summer Camp) Main Organizer, lecturer, and problem setter for the algorithm course in Iran National Olympiad in Informatics 2021. Delivered lectures on flow algorithms, number theory, and dynamic programming. Designed 3 out of 9 final exam problems.	Jul. 2021 - Aug. 2021
Teaching Assistant for Natural Language and Computation (MIT 6.S051, Prof. Robert Berwick) Revised and created new lab practices including: Segmentation, Parsers, Semantic Parsing with Lambda Calculus, and Grammar Inference.	Sep. 2022 - Dec. 2022
Workshop Organizer (MIT Global Teaching Lab, Armenia) Organized a month-long workshop on algorithmic problem-solving for high school students in Armenia to foster enthusiasm for computer science. The workshop began with 10 students and concluded with 40.	Jan. 2024 - Jan. 2024